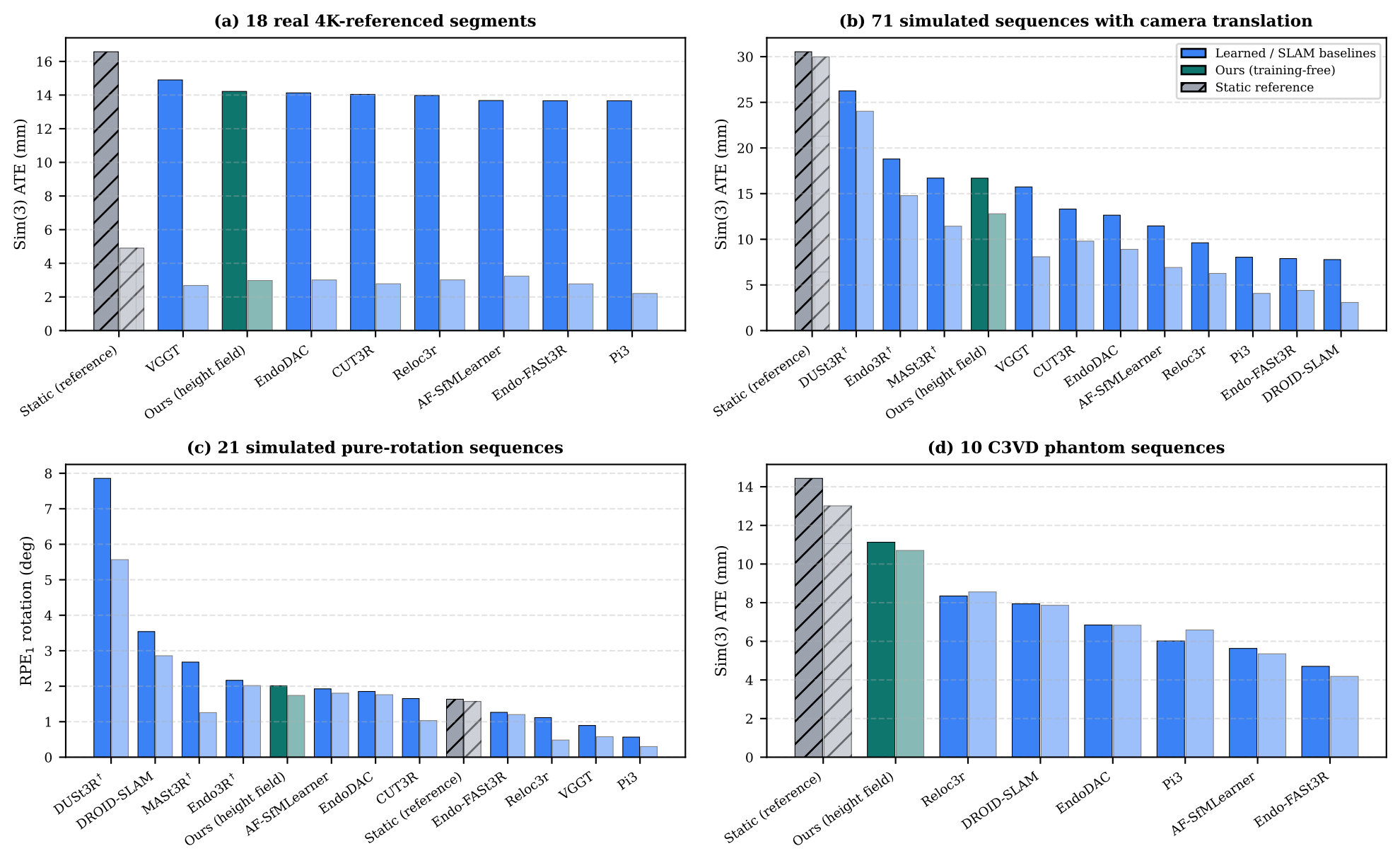
(a) 18 real 4K-referenced segments
Sim(3) ATE (mm)
Static (reference)
VGGT
Ours (height field)
EndoDAC
CUT3R
Reloc3r
AF-SfMLearner
Endo-FASt3R
Pi3
(b) 71 simulated sequences with camera translation
Learned / SLAM baselines
Ours (training-free)
Static reference
Sim(3) ATE (mm)
Static (reference)
DUSt3R†
Endo3R†
MASt3R†
Ours (height field)
VGGT
CUT3R
EndoDAC
AF-SfMLearner
Reloc3r
Pi3
Endo-FASt3R
DROID-SLAM
(c) 21 simulated pure-rotation sequences
RPE1 rotation (deg)
DUSt3R†
DROID-SLAM
MASt3R†
Endo3R†
Ours (height field)
AF-SfMLearner
EndoDAC
CUT3R
Static (reference)
Endo-FASt3R
Reloc3r
VGGT
Pi3
(d) 10 C3VD phantom sequences
Sim(3) ATE (mm)
Static (reference)
Ours (height field)
Reloc3r
DROID-SLAM
EndoDAC
Pi3
AF-SfMLearner
Endo-FASt3R